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From Search Intent to Retrieval Demand: A Pre-Generation Framework for Generative Engine Optimization (GEO)

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Abstract

The shift from classical Information Retrieval (IR) toward Generative Engines (GE) necessitates a re-evaluation of user intent. Where search engines served as deterministic routers, GEs act as probabilistic answer machines—rendering established taxonomies such as Broder's "Navigational/Informational/Transactional" model insufficient for predicting LLM behavior. This position paper proposes the Generative Intent Operationalization (GIO), a predictive, pre-generation framework that classifies user intent before the model produces output—unlike existing approaches that analyze interaction traces post hoc. GIO classifies intent based on Grounding Necessity (GN): the normative epistemic requirement for a model to retrieve external evidence—what a system should retrieve to adequately fulfill the user's need. Synthesizing findings from agentic architectures—the epistemic necessity principle (Wang et al., 2025), Self-RAG (Asai et al., 2024), and FLARE (Jiang et al., 2023)—we argue that GN governs visibility in AI-generated answers. This makes GIO directly actionable for Generative Engine Optimization (GEO): rather than retroactively explaining which content appeared, GIO enables strategists to anticipate retrieval demand and optimize proactively. Empirical validation is ongoing; this paper presents the theoretical framework, its architectural basis, and a falsifiable validation protocol.

Keywords

computing and processing, epistemic uncertainty, generative engine optimization, grounding necessity, large language models, search intent

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This position paper proposes the Generative Intent Operationalization (GIO), a *predictive, pre-generation* framework that classifies user intent *before* the model produces output—unlike existing approaches that analyze interaction traces *post hoc*. GIO classifies intent based on Grounding Necessity (GN): the *normative* epistemic requirement for a model to retrieve external evidence—what a system *should* retrieve to adequately fulfill the user's need. Synthesizing findings from agentic architectures—the epistemic necessity principle (Wang et al., 2025), Self-RAG (Asai et al., 2024), and FLARE (Jiang et al., 2023)—we argue that GN governs visibility in AI-generated answers. This makes GIO directly actionable for *Generative Engine Optimization* (GEO): rather than retroactively explaining which content appeared, GIO enables strategists to anticipate retrieval demand and optimize proactively. Empirical validation is ongoing; this paper presents the theoretical framework, its architectural basis, and a falsifiable validation protocol.

***Index Terms*—Generative Engine Optimization, Grounding Necessity, Epistemic Uncertainty, Search Intent, Retrieval-Augmented Generation, Large Language Models**

I. Introduction: The Epistemic Shift

The transformation of Information Retrieval (IR) systems into Generative Engines (GE) marks a significant structural shift in human-machine interaction. While classical web search taxonomies [3], [10] modeled user needs in terms of reaching destinations, finding information, or performing transactions, GEs function as probabilistic answer machines. This transformation was anticipated by Metzler et al. [15], who envisioned hybrid systems that generate expert-level answers while retaining references to supporting documents.

This shift fundamentally alters the nature of "User Intent." The user no longer primarily seeks a location, but a synthesis. For the content strategist, this creates a critical opacity: the "Black Box" of generation. Does the answer stem from the model's parametric memory (*Memorization*), or was it constructed through external retrieval (*Grounding*)?

This paper identifies the central deficit: We lack a framework to predict *Grounding Necessity* based on the prompt alone. We propose the **Generative Intent Operationalization (GIO)**, a taxonomic framework aligning user goals with the engine's retrieval requirements as determined by epistemic constraints. GIO serves as the theoretical superstructure for Generative Engine Optimization (GEO).

To clarify the contribution:

- **Not a System Architecture:** GIO does not propose a new retrieval algorithm or vector database structure.
- **Not a Post-Hoc Analysis:** GIO models intent *a priori*—before the model generates a response.
- **Not a Prompt Engineering Guide:** GIO focuses on the informational nature of the request, not syntactical optimization.

Instead, GIO contributes the first *pre-generation operationalization* that classifies user intent based on probabilistic grounding necessity—independent of the specific model architecture. GIO classifies based on the epistemic demand inherent in the prompt—what a system *should* retrieve—rather than what current systems happen to retrieve.

To illustrate GIO's practical value, consider the query *"What are the new EU AI Act compliance requirements for generative AI providers in 2026?"* Under Broder's taxonomy [3], this is "Informational"—a label that tells a content strategist nothing actionable. GIO classifies it as **Mode 1.3 Advisory** with **High Grounding Necessity**: the model cannot answer from parametric memory alone, because the regulatory landscape is volatile (high $V_{volatility}$), the query targets a specific legal instrument (high E_{spec}), the legislation postdates most training cutoffs (high T_{decay}), and the prompt implicitly demands current expert analysis (high I_{gap}).

II. Foundation: Generative Engine Optimization

The literature on GEO has grown rapidly between 2024 and 2026, distinguishing itself from traditional SEO. Aggarwal et al. [1] provide empirical proof that visibility in generative answers can be manipulated, demonstrating that optimization tactics—citations, quotations, statistical data—can increase source visibility by up to 40%. However, Aggarwal focuses on *tactics* (what to change). GIO focuses on *strategy* (when to optimize).

A critical distinction separates GEO from Answer Engine Optimization (AEO). AEO targets "Featured Snippets" through literal extraction of a sentence (GIO Mode 1.1). GEO targets the inclusion of a document as evidence in a newly generated argument. Bagga et al.'s "E-GEO" [2] demonstrates this in e-commerce, showing that product description optimization can measurably increase source inclusion in generative answers.

III. Alternative Taxonomies

GIO competes directly with recent classification models for interpretive authority.

TUNA (Shelby, Diaz, & Prabhakaran, 2025) [5] proposes 6 modes and 57 query types, including social categories. While descriptively excellent for HCI research, it is operationally too granular for optimization. GIO collapses TUNA's social categories into "Low Grounding Necessity," rendering them non-actionable for GEO.

NBER (Chatterji et al., 2025) [4] identifies "Writing" and "Seeking Information" as dominant usage patterns. The category "Writing" is dangerously ambiguous: it conflates *Ungrounded Generation* ("Write a cover letter") with *Grounded Generation* ("Summarize the latest ECB monetary policy report"). GIO splits this category.

Automated Taxonomies (Shah et al., 2024) [6] propose LLM-generated taxonomies. While topics change dynamically, the epistemic nature of a query remains invariant. GIO is a robust *meta-framework* that operates at the epistemic level.

IV. Architectural Basis of Grounding

GIO postulates Grounding Necessity as an epistemic property of the prompt. This is *corroborated* by recent literature which treats GN as a hard architectural constraint.

Wang et al. (2025) in *Theory of Agent* [7] argue that an agent should invoke external tools *only* when internal reasoning is insufficient to resolve epistemic uncertainty. GIO describes the publisher's strategy for the exact phenomenon that ToA describes as the agent's policy.

Technical implementations confirm GIO's relevance:

- **Self-RAG** [8] trains models to generate reflection tokens (e.g., `[Retrieve=yes]`)—the algorithmic implementation of high GN.

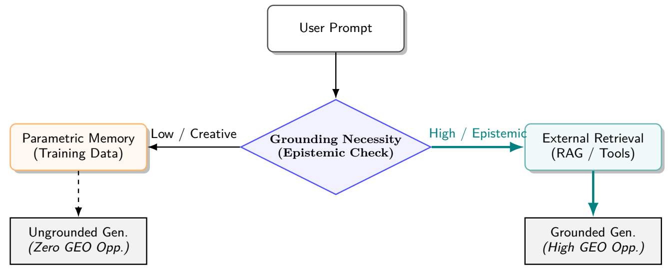

Fig. 1. The GIO Decision Flow. High Grounding Necessity triggers the retrieval path, creating the GEO opportunity.

- **FLARE** [9] triggers retrieval when token-level confidence drops below a threshold, providing a fine-grained operationalization of grounding necessity.
- **Adaptive-RAG** [13] uses a classifier to route queries into no-retrieval, single-step, and multi-step retrieval—mapping onto GIO's Intent-Complexity Nexus.
- **Dhole** (2025) [14] employs output-level uncertainty metrics to dynamically decide whether retrieval should be triggered mid-generation.

A crucial demarcation applies: these systems address the "retrieve or not" question *for the system architect*—optimizing when the engine should trigger retrieval *during* generation. GIO addresses the complementary question *for the content strategist*, who must assess *before* generation whether a query is likely to activate the retrieval path. Early work on retrieval-augmented generation, notably WebGPT [16], already demonstrated that LLMs can learn to issue search queries when parametric knowledge is insufficient.

GIO defines *normative* Grounding Necessity—what a system *should* retrieve—rather than *descriptive* GN—what current systems *happen to* retrieve. The convergence toward normative retrieval is structurally predictable: systems that fail to retrieve when epistemic necessity is high produce hallucinated or stale answers, leading to user dissatisfaction. Market pressure therefore drives retrieval calibration toward the normative target that GIO defines.

V. The Generative Intent Operationalization

GIO classifies prompts based on the probability that a valid answer requires external data access (see Figure 1).

A. The Intent-Complexity Nexus

GIO must be mapped against **Structural Prompt Complexity**. A simple query often yields a parametric answer, while a complex Chain-of-Thought prompt forces fact verification. By intersecting Intent and Complexity, we identify "Zones of Opportunity" (Table II).

The matrix reveals that Grounding Necessity increases with prompt complexity across all modes. "What is the EU AI Act?" falls into the *Zone of Parametric Response*. The full query from Section I occupies the *Zone of Synthesis*, demanding comparison across external sources.

TABLE I

The Generative Intent Operationalization (GIO) — Classification Matrix

Mode	Category	User Intent & Context	Grounding Necessity & GEO Strategy
1. ASKING	**1.1 Fact Retrieval**	**Intent:** Verify a static fact (“Height of Eiffel Tower”). *Constraint: High Consensus.*	**Low.** The Parametric Trap. *Strategy: Entity Authority (Zero-Click).*
	1.2 Real-Time Synthesis	**Intent:** Access dynamic data (“Stock price”, “Election results”). *Constraint: High Volatility.*	**High.** Requires structured data retrieval. *Strategy: Schema Markup & Data Feeds.*
	1.3 Advisory	**Intent:** Seek expert perspective (“EU AI Act compliance”). *Constraint: Subjective Nuance.*	**High.** Requires multi-perspective synthesis. *Strategy: Thought Leadership.*
2. DOING	**2.1 Utility**	**Intent:** Transformation of existing input (“Translate to Spanish”).	**None.** Tool/function execution. *GEO Strategy: N/A.*
	2.2 Ungrounded Gen.	**Intent:** Creative creation ex nihilo (“Write a poem”).	**Low.** Relies on internal creativity. *Strategy: Style Prompting.*
	2.3 Grounded Gen.	**Intent:** Source-dependent creation (“Summarize this PDF”). *Constraint: User-provided source.*	**Grounding from Context.** Source provided by user. *Strategy: In-Document Optimization.*
3. ACTING	**3.1 Transactional**	**Intent:** Execute a purchase or booking (“Book a flight”).	**High.** API dependent. *Strategy: Agent Actions.*
	3.2 Open-Ended Investigation	**Intent:** Solve an underspecified problem (“Plan a trip to Japan”).	**High.** Iterative retrieval required. *Strategy: Deep Content & Hubs.*

TABLE II

The Unified GEO Matrix — Intersection of Intent (GIO) and Complexity

Prompt Complexity → GIO Intent ↓	**Level 1: Zero-Shot**	**Level 2: Context-Rich**	**Level 3: Reasoning**
Mode 1: ASKING	*Zone of Parametric Response* GEO Value: None	*Zone of Refinement* GEO Value: Low	*Zone of Synthesis* GEO Value: High
Mode 2: DOING	*Zone of Creativity* GEO Value: None	*Zone of Templating* GEO Value: Low	*Zone of Grounding* GEO Value: Maximal
Mode 3: ACTING	*Zone of Delegation* GEO Value: Low	*Zone of Orchestration* GEO Value: Medium	*Zone of Autonomy* GEO Value: High

B. The Parametric Trap

A central concept in GIO is the *Parametric Trap*: the condition in which a model generates a confident but epistemically invalid response because the query *appears* answerable from parametric memory. The trap is sprung when high entity specificity (E_{spec}) in the training corpus masks high volatility ($V_{volatility}$). The model's confidence is miscalibrated by familiarity.

The empirical basis is established by Mallen et al. [17], who demonstrate that entity popularity in training data strongly predicts answer correctness. Their “Adaptive Retrieval” method skips retrieval for popular entities—exposing the Parametric Trap for popular entities whose facts have changed.

C. The Grounding Necessity Hypothesis

We propose that Grounding Necessity is governed by four variables whose structural relationships can be stated as falsifiable qualitative claims. The need for multi-dimensional routing is empirically supported by Wang et al. [18], whose evaluation across 7,727 queries demon-

TABLE III
THE PARAMETRIC TRAP ACROSS DOMAINS.

Query	E_{sp}	V_{vol}	**Trap**	**Mechanism**
"Height of the Eiffel Tower"	High	≈ 0	No	Immutable fact
"Capital of France"	High	≈ 0	No	Geopolitical constant
"CEO of Google"	High	High	Yes	Head entity masks volatile role
"Best treatment for Type 2 diabetes"	Mod.	High	Yes	Medical consensus evolves

strates that no single RAG paradigm is universally optimal.

We propose **Hypothesis H1** (Structural Hypothesis) as three falsifiable claims:

H1-i **Gating:** I_{gap} (Information Gap) acts as a *necessary condition* for GN. When the prompt signals no information-seeking intent, GN remains low regardless of topic properties.

H1-ii **Amplification:** T_{decay} and $E_{spec} \times V_{volatility}$ act as *amplifiers* that increase GN, but only when the gate is open (I_{gap}: High).

H1-iii **Interaction:** E_{spec} and $V_{volatility}$ interact *jointly*. Entity specificity drives GN only when the entity is subject to change—resolving the Specificity Paradox.

The four variables are:

- I_{gap}: **Information Gap**—the degree to which the prompt signals a deficit that parametric memory cannot fill. Adapted from Loewenstein's information gap theory [11]. Crucially, I_{gap} resists reliable algorithmic detection due to conversational implicature.
- T_{decay}: **Temporal Decay**—the distance from the model's training cutoff to the referenced event.
- E_{spec}: **Entity Specificity**—measurable via NER pipelines and knowledge graph lookup.
- $V_{volatility}$: **Volatility**—ranging from immutable (gravitational constant) to real-time (stock prices). Measurable via Wikipedia revision frequency.

Table IV operationalizes H1 as an ordinal decision framework.

The gating structure (H1-i) makes a precise exclusion: **volatility and temporal decay cannot generate GN in the absence of information-seeking intent.** A creative prompt about a volatile entity ("Write a limerick about the stock market") has maximal $V_{volatility}$ but low I_{gap}, yielding low GN. The principal vulnerability lies in prompts with *implicit* information demands through conversational implicature—a central open question for future automation of the framework.

TABLE IV
ORDINAL OPERATIONALIZATION OF H1.

I_{gap}	T_{dec}	$E_{sp} \times V_{vol}$	GN	**GEO Action**
Low	Any	Any	Low	Monitor
High	Low	Low	Low	Structural Opt.
High	High	Low	Med	Freshness Signaling
High	Low	High	Med	Domain Authority
High	High	High	High	Full GEO

VI. DISCUSSION AND LIMITATIONS

Personalization Wall. GIO models "Public Intent." RAG systems increasingly utilize personalized context (email, calendar). A query like "When is my flight?" has maximal GN but zero GEO opportunity for public publishers. GIO's strategies apply only to the *public retrieval* path. However, GIO's epistemic classification is equally valid within private RAG systems such as enterprise knowledge management.

Epistemic Lock-in. If engines exclusively retrieve from optimized High-GN sources, a feedback loop may marginalize non-optimized but high-quality content. Source diversity metrics in retrieval pipelines could mitigate this risk.

Single-Turn Scope. GIO currently models individual prompts. Grounding Necessity frequently emerges during multi-turn conversations. Extending GIO to model GN as a function of conversational state is a necessary next step.

Validity of GN Hypothesis. The structural hypothesis deliberately omits a quantitative functional form. I_{gap}—the gating variable—resists reliable algorithmic measurement. The ordinal operationalization (Table IV) provides a usable framework; conversion into a continuous model requires advances in detecting implicit information demands.

VII. PLANNED VALIDATION

A validation study is in preparation. We will sample $N = 1{,}000$ prompts from the WildChat corpus [12] using stratified sampling. Three domain experts will annotate prompts using the GIO classification matrix, targeting Fleiss' $\kappa > 0.7$. The GIO model will be compared against a keyword-matching baseline and a trained BERT classifier using paired McNemar's tests. A pilot annotation study ($N = 55$) has been completed; the filtered prompt pool (230,289 prompts) is publicly available [19], and replication materials are available on GitHub [20].

VIII. CONCLUSION

This paper argued that the transition from search engines to generative engines demands a new organizing principle for user intent: Grounding Necessity—the normative epistemic requirement for a system to retrieve external evidence. The resulting framework, GIO, classifies intent *before* generation, produces falsifiable predictions through its gating structure (I_{gap} as necessary condition), and makes the epistemic demand of a prompt directly actionable for Generative Engine Optimization. The critical next step is empirical validation: determining whether GIO's structural commitments add predictive value beyond the raw features they organize. If validated, GIO reframes GEO from content optimization to demand shaping.

Author Contributions

K. Spriestersbach conceived the framework, conducted the analysis, and wrote the manuscript. S. Vollmer supervised the research and provided critical revisions.

References

[1] P. Aggarwal et al., "GEO: Generative Engine Optimization," in *Proc. 30th ACM SIGKDD*, 2024. arXiv:2311.09735.

[2] S. Bagga et al., "E-GEO: A Testbed for Generative Engine Optimization in E-Commerce," 2025. arXiv:2511.20867.

[3] A. Broder, "A taxonomy of web search," *SIGIR Forum*, vol. 36, no. 2, pp. 3–10, 2002.

[4] A. Chatterji et al., "How People Use ChatGPT," NBER Working Paper 34255, 2025.

[5] R. Shelby, F. Diaz, and V. Prabhakaran, "TUNA: A Taxonomy of User Needs and Actions for LLMs," 2025. arXiv:2510.06124.

[6] C. Shah et al., "Using Large Language Models to Generate, Validate, and Apply User Intent Taxonomies," 2024. arXiv:2309.13063.

[7] B. Wang et al., "Position: Agent Should Invoke External Tools ONLY When Epistemically Necessary," 2025. arXiv:2506.00886.

[8] A. Asai, Z. Wu, Y. Wang, A. Sil, and H. Hajishirzi, "Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection," in *Proc. ICLR*, 2024. arXiv:2310.11511.

[9] Z. Jiang et al., "Active Retrieval Augmented Generation," in *Proc. EMNLP*, pp. 7969–7992, 2023. arXiv:2305.06983.

[10] D. E. Rose and D. Levinson, "Understanding user goals in web search," in *Proc. WWW '04*, pp. 13–19, 2004.

[11] G. Loewenstein, "The psychology of curiosity: A review and reinterpretation," *Psychological Bulletin*, vol. 116, no. 1, pp. 75–98, 1994.

[12] W. Zhao et al., "WildChat: 1M ChatGPT Interaction Logs in the Wild," 2024. arXiv:2405.01470.

[13] S. Jeong et al., "Adaptive-RAG: Learning to Adapt Retrieval-Augmented Large Language Models through Question Complexity," in *Proc. NAACL*, 2024. arXiv:2403.14403.

[14] K. D. Dhole, "To Retrieve or Not to Retrieve? Uncertainty Detection for Dynamic Retrieval Augmented Generation," 2025. arXiv:2501.09292.

[15] D. Metzler, Y. Tay, D. Bahri, and M. Najork, "Rethinking Search: Making Domain Experts out of Dilettantes," *SIGIR Forum*, vol. 55, no. 1, pp. 1–27, 2021.

[16] R. Nakano et al., "WebGPT: Browser-Assisted Question-Answering with Human Feedback," 2021. arXiv:2112.09332.

[17] A. Mallen et al., "When Not to Trust Language Models: Investigating Effectiveness of Parametric and Non-Parametric Memories," in *Proc. ACL*, 2023.

[18] Z. Wang et al., "RAGRouter-Bench: A Dataset and Benchmark for Adaptive RAG Routing," 2026. arXiv:2602.00296.

[19] K. Spriestersbach, "WildChat-GIO: Filtered English/German Prompt Pool for the GIO Pilot Annotation Study," Zenodo, 2026. https://doi.org/10.5281/zenodo.18593414

[20] K. Spriestersbach, "GIO Pilot Study — Replication Materials," GitHub, 2026. https://github.com/kaispriestersbach/gio-pilot-study

[21] S. Yan et al., "Corrective Retrieval Augmented Generation," 2024. arXiv:2401.15884.